

Segmentation fault when using VQE with COBYLA optimizer

<https://github.com/qiskit-community/qiskit-aqua/issues/80>

ROW 64

Segmentation fault when using VQE with COBYLA optimizer #80



Closed



yaelbh opened on Aug 9, 2018

...

Informations

- Qiskit Aqua version: current master
- Python version: 3.6
- Operating system: Linux

What is the current behavior?

A segmentation fault occurs for a specific input.

Steps to reproduce the problem

Create the following input file of aqua chemistry:

=====

```
&name
LiH excited states molecule experiment. Var for SWAPRZ with VQE
&end
```

```
&driver
name=PYSCF
&end
```

```
&problem
name=energy
enable_substitutions=True
random_seed=50
&end
```

```
&pyscf
atom=H 0 0 -2.5; Be 0 0 0; H 0 0 2.5
unit=Angstrom
charge=0
spin=0
basis=sto3g
&end
```

```
&operator
name=hamiltonian
transformation=particle_hole
qubit_mapping=jordan_wigner
two_qubit_reduction=False
freeze_core=True
orbital_reduction=[]
max_workers=4
&end
```

```
&algorithm
name=VQE
operator_mode=matrix
initial_point=None
&end
```

```
&optimizer
name=COBYLA
maxiter=10000
&end
```

```
&variational_form
name=SWAPRZ
depth=1000
entangling=full
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

Customize

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Participants



Visual Studio

```
&variational_form
name=SWAPRZ
depth=1000
entanglement=full
entangler_map=None
&end

&initial_state
name=HartreeFock
qubit_mapping=jordan_wigner
two_qubit_reduction=False
&end

&backend
name=local_statevector_simulator
&end

=====

Then run:
qiskit_aqua_chemistry_cmd <file name>
```

The input cam from Julia Rice, but with other values of depth and maxlter. The bug occurs for these specific depth and maxlter (depth=1000, maxlter=10000). I haven't witnessed it for other values of these parameters.

What is the expected behavior?

Run without segmentation fault.

Suggested solutions

For the moment I only know that it happens inside COBYLA.optimize.



yaelbh on Aug 9, 2018

Contributor Author ...

After the call to `scipy.optimize`, the segmentation fault happens before the call to `vqe._energy_evaluation`



chunfuchen on Aug 9, 2018

Contributor ...

I think this is the problem of COBYLA optimizer. With `depth=1000`, you will have 78000 parameters. It seems that it exceeds what COBYLA can do.

Here is a simple test about COBYLA (not involving aqua)

```
from scipy.optimize import rosen, minimize
import numpy as np

x0 = np.random.random(78000) # number of paramters
res = minimize(rosen, x0, tol=1e-06, method="COBYLA")
```



It raises segmentation error as well.

I suggest to reduce the depth size or change the optimizer.



yaelbh on Aug 12, 2018

Contributor Author ...

Great, thanks, closing the issue then.



yaelbh closed this as **completed** on Aug 12, 2018